

Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738153410910-Chapter 9 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

HOW TECHNOLOGY HELPS IN FLOOD PREVENTION

In this project, you will design and build a Smart Watering Device that can help conserve water and ensure your plants are always properly hydrated. The device will sense when your plants need watering and automatically provide water when necessary. You'll need to think about how to monitor soil moisture, control water delivery, and provide feedback about the system's status.

HOW TECHNOLOGY HELPS IN WATERING AND CONSERVATION

Technology allows us to care for plants more efficiently by using sensors and automation to deliver just the right amount of water. This helps prevent overwatering or underwatering, both of which can be harmful to plants. By creating a system that responds to the environment, we can make sure our plants stay healthy while conserving water.

IDEA GENERATION

For this project, you'll need to think carefully about the components required to build a Smart Watering Device. Below are some hints to guide you in choosing the right modules:

1. Monitoring Soil Moisture:

- Consider what kind of sensor could detect when the soil is dry. This sensor will need to be placed in the soil to constantly monitor moisture levels. When the soil gets too dry, it will need to trigger a response.
- Hint: Think about a sensor that can measure environmental factors, such as moisture in the soil.



2. Automating the Watering Process:

- You'll need a way to deliver water to the plants when the sensor detects that the soil is too dry. Think about what kind of mechanism could turn on and off to control water flow.
- Hint: Consider using a motor that could be connected to a pump or another mechanism to move water.



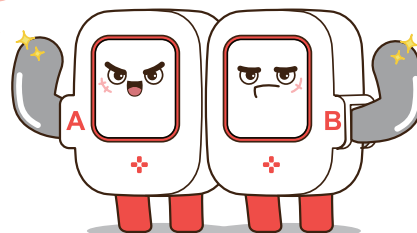
2. Automating the Watering Process:

- The system should give you some sort of indication when it's watering your plants. Think about how you could use lights or a display to show the system's status or to alert you when the watering is taking place.
- Hint: Think about components that can show text or light up to provide real-time feedback on what the system is doing.



HINT

Think about the role of each component in your Smart Watering Device. The sensor should detect changes in the environment, the motor should control the flow of water, and the display or lights should inform you of the system's status. When brainstorming, ask yourself: "What function does each part need to perform?"



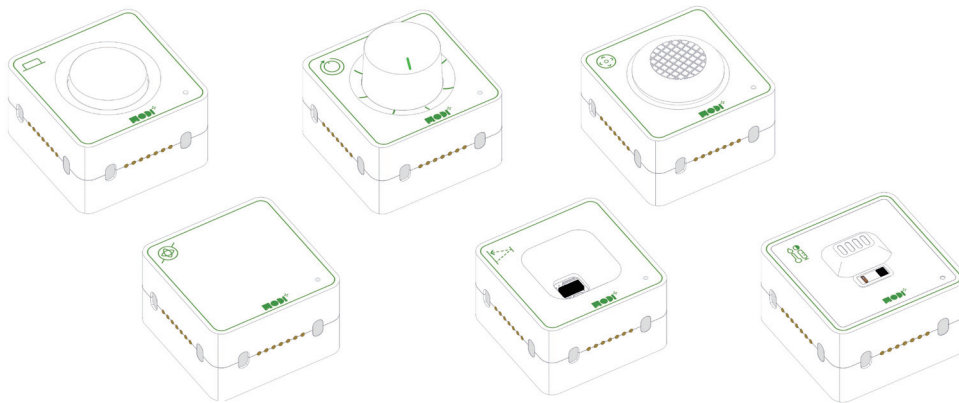


Required Modules & Brainstorming

To build our Flood Alarm System, we need to decide which MODI modules to use. Here are some ideas to get started:

SENSOR MODULES

Detecting Soil or Environmental Conditions: Think about a sensor that could monitor the moisture level of the soil or the surrounding air. This sensor might measure variables like humidity, temperature, or other environmental factors to decide when watering is needed.

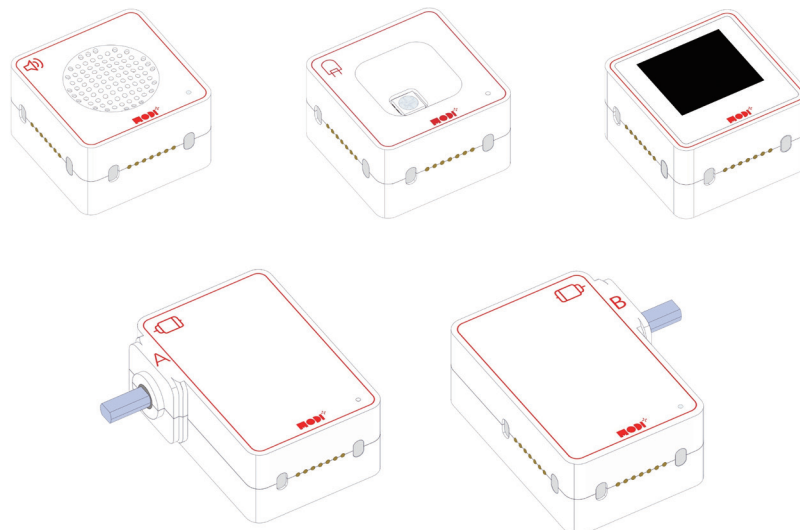


OUTPUT MODULES

Activating the Watering Process: Consider a mechanism that can physically water the plants. You might need a module that can control a movement to release water.

Providing Visual Feedback: Reflect on how you might want to see the status of the watering system. Think about a way to display information, such as when the system is watering or what the current moisture level is.

Alerting You to the System's Actions: Imagine a way the system could alert you when it's operating. This could be through lights, sounds, or even a display of messages.



Required Modules & Brainstorming

Which MODI modules do you need for the creation?



BUTTON



DIAL



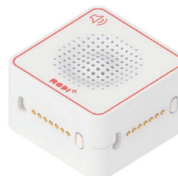
ENVIRONMENT



JOYSTICK



TOF



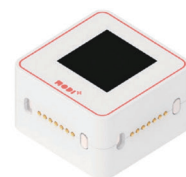
SPEAKER



IMU



LED



DISPLAY



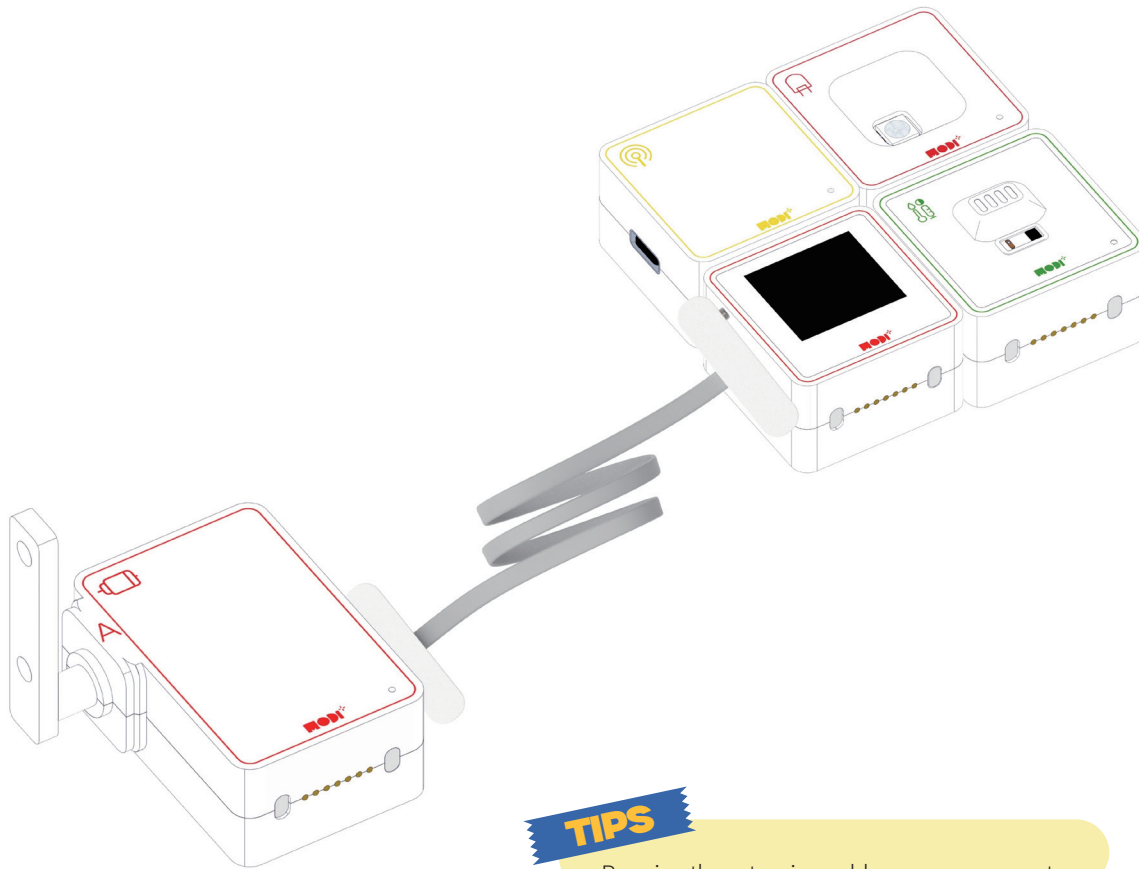
MOTOR A



MOTOR B

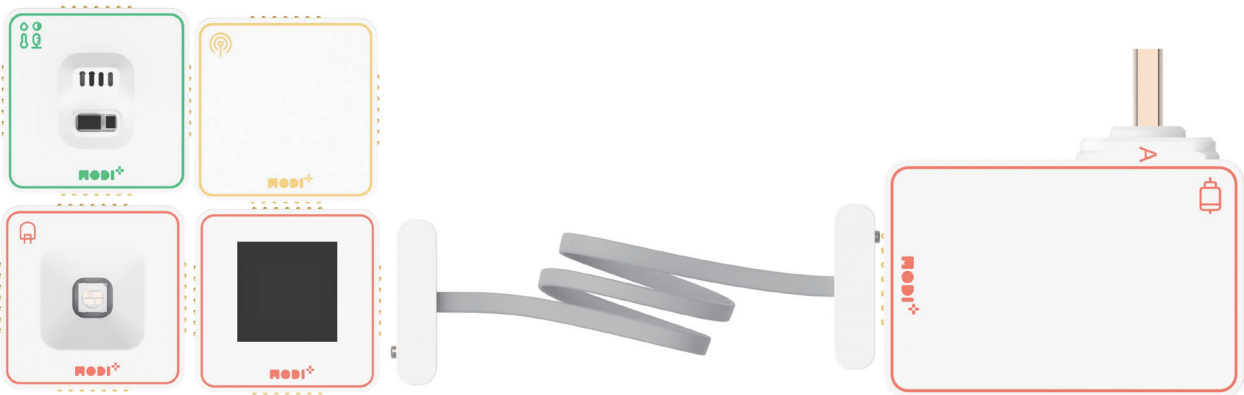
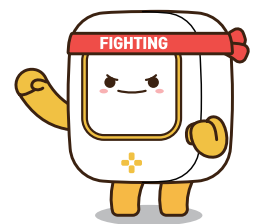


WHEN WE COMBINE ALL THE MODULES, THE CREATION HAS TO BE LOOK LIKE THIS.



TIPS

By using the extension cable, you can connect modules even if they are far apart from each other.



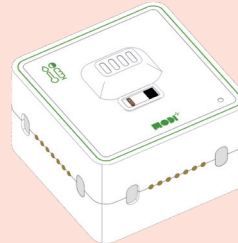
Module Functionality

ENVIRONMENT MODULE

How It Works: The Environment module can measure various environmental factors, such as temperature, humidity, and sound. For this smart watering system, it will be primarily used to detect soil moisture levels or surrounding humidity to determine when the plants need watering.

Applications in Smart Watering System

- **Humidity Monitoring:** The module can monitor ambient humidity, helping to adjust the watering schedule based on the environment's dryness.



DISPLAY MODULE

How It Works: The Display module shows text or graphical information, providing users with updates on the system's status and the environment.

Applications in Smart Watering System

- **Alert Messages:** When the moisture level is too low, the Display can show a message such as "Watering Needed" to notify users of the system's actions.

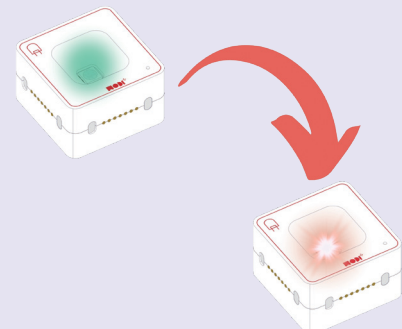


DISPLAY MODULE

How It Works: The LED module is used to provide visual signals. It lights up in different colors to indicate various levels of alerts.

Applications in Smart Watering System

- **Watering Indicator:** The LED lights up in a specific color (e.g., green) when the system is actively watering the plants.
- **Alert Indicator:** If the soil moisture is critically low, the LED could change to a different color (e.g., red) to signal an urgent need for water.



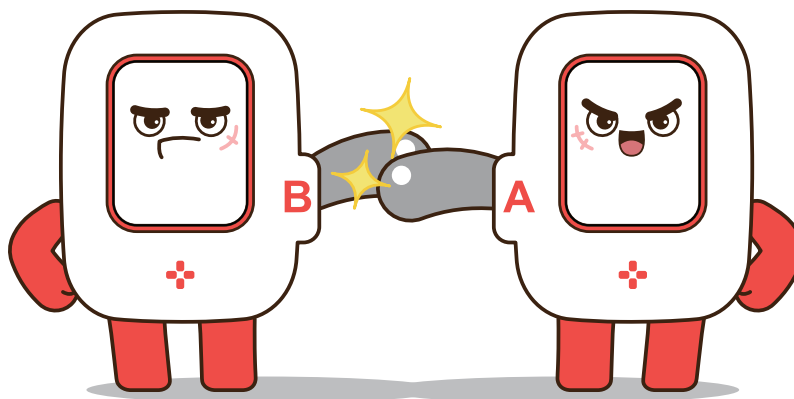
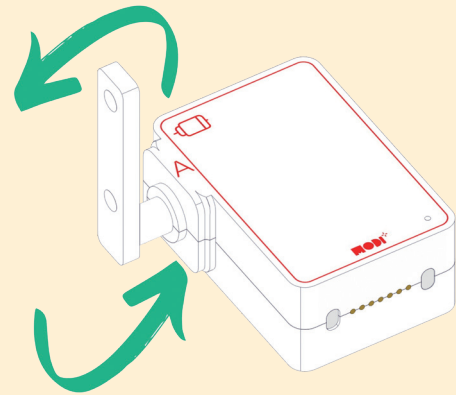


MOTOR A MODULE

How It Works: The Motor A module is responsible for mechanical movement. In this smart watering system, it will control the valve or pump that dispenses water to the plants.

Applications in Smart Watering System

- **Water Dispensing:** The Motor A module activates to open a valve or power a pump, allowing water to flow to the plants when needed.
- **Controlled Water Flow:** The motor ensures that water is dispensed efficiently, preventing over-watering or under-watering based on the soil's moisture level.



WHY THESE MODULES ARE EFFECTIVE IN A SMART WATERING SYSTEM

The environment sensor is used to monitor soil moisture levels or environmental conditions, while the display and LED modules provide visual feedback. Here are some ways you might find these modules useful:

Comprehensive Monitoring

By combining environmental sensing (like moisture levels), visual feedback (Display and LED), and action mechanisms (such as Motor A for watering), the system ensures that plants receive the right amount of water. For instance, the display might show moisture levels, while the LED indicates when watering is needed.



DISCUSSION POINT



Think about how monitoring systems in smart homes or gardens work. What other parameters might be important to measure along with moisture, and how could they be integrated?

Redundancy

Having both visual (Display and LED) and action components (Motor A) ensures that if one feedback mechanism (like the LED) is missed, the system still waters the plants or provides the necessary visual data on the display.



DISCUSSION POINT



Why is it essential to have multiple indicators in a smart system? Can you think of a situation where one method might fail, and how the other would ensure the system still works effectively?